Instructions

In this assignment you will build recommender systems to make predictions related to book reviews from *Goodreads.*

Submissions will take the form of prediction files uploaded to gradescope, where their test set performance will be evaluated on a leaderboard. **Most of your grade will be determined by 'absolute' cutoffs; a smaller fraction of your grade will be based on your leaderboard score.** The absolute cutoffs will be set to be (reasonably) achievable; the leaderboard component is to ensure that there's a distribution of grades, though should not prevent anyone from being able to pass the assignment.

The assignment is due at the beginning of week 8, though **the exact deadline is as posted on gradescope.** Make sure you upload solutions to the leaderboard regularly in order to make continual progress.

You should submit two files:

- `writeup.txt` a brief, plain-text description of your solutions to each task; please prepare this adequately in advance of the submission deadline; this is only intended to help us follow your code and does not need to be detailed.
- `assignment1.py` A python file containing working code for your solutions. *The autograder will not execute your code;* this file is required so that we can assign partial grades in the event of incorrect solutions, check for plagiarism, etc. Your solution should clearly document which sections correspond to each task. We may occasionally run code to confirm that your outputs match submitted answers, so please ensure that your code generates the submitted answers.

Along with three files corresponding to your predictions:
`task_1_predictions.csv ... task_6_predictions.csv`
These files should contain your predictions for each (test) instance. **The provided baseline code demonstrates how to generate valid output files.**

You are required to submit something for at least 3 of the 6 tasks. If you submit to more than 3 tasks, your best 3 scores will be kept. Each task will be worth 10 points.

This assignment should be completed individually. To begin, download the files for this assignment from:
https://drive.google.com/file/d/1Gu0TAAFSPrS6ZwT1K0D7ADf1lhtlcqJg/view?usp=sharing

Files

- `tasks_1_2_train.csv` ratings/interactions to be used for tasks 1 and 2. The files for tasks 3-6 are similar, and are just based on different samples of the original dataset. Each row has `user, item, timestamp,` and `rating` fields.

- `task_X_test.csv`:
    - user : the user
    - item : the item
    - item1 / item2 : a pair of items, for tasks where you have to guess which is the "positive" interaction
    - timestamp : the time of the interaction

  There are also fields that must be predicted, which always go in the final colum:
    - rating : a rating
    - which : which among item1 or item2 is the "positive" instance. You should fill in one of the two item IDs
    - delta : a time delta (difference between timestamps)

Please do not try to collect these reviews from Goodreads, or to reverse-engineer the hashing function I used to anonymize the data. Doing so will not be easier than successfully completing the assignment. **We may execute code for any solution suspected of violating the competition rules to confirm that it generates valid output;** code will be run through a plagiarism detector. **We may assign a zero grade if your code does not appear to correspond to your submitted solution.**

Tasks

You are expected to complete (at least three of) the following six tasks:

1. **Rating prediction:** Predict a rating given a user / item pair
2. **Interaction prediction:** Predict which of a pair of interactions is the positive one
3. **Rating prediction:** Same as above, but the timestamp is available
4. Interaction prediction: Same as above, but the timestamp is available
5. **Interaction prediction:** Same as above, but items are ordered sequentially, and the prediction is always concerned with the user's most recent interaction (i.e., you must predict what item comes next). The timestamp is not available in the test set, so you may only use the ordering to improve your predictions.
6. **Time delta prediction:** Predict the amount of time that will elapse before the user's next interaction

A competition page has been set up on gradescope to keep track of your results compared to those of other members of the class. The leaderboard will show your results on half of the test data, but your ultimate score will depend on your predictions across the whole dataset.

Grading and Evaluation

This assignment is worth 33 marks. You will be graded on the following aspects. Each of the three tasks is worth 10 marks, plus 3 marks for the written report. Your grade for each task will be based on:

- Your ability to obtain a solution which outperforms the baselines on the *unseen portion of* the test data (4 marks for each task). Obtaining full marks requires a solution which is substantially better than baseline performance.
- Obtain a solution which outperforms the baselines on the *seen portion of* the test data. This is a sort of consolation prize in case you overfit to the seen portion (2 marks for each task).
- Your ranking for each of the tasks compared to other students in the class (4 marks for each task).

Note that there is more emphasis on the ranking based component this year (4 marks per task). **But,** your final score will be based on your three best task scores, so you can do well on the assignment as long as you do well on *some* of the tasks.

Finally, your written report should describe the approaches you took to each of the tasks. Your report should contain a few sentences describing your solution to each task. You will generally get full marks for the report automatically unless some issue is suspected with your solution.

To obtain good performance, you should not need to invent new approaches (though you are more than welcome to!) but rather you will be graded based on your decision to apply reasonable approaches to each of the given tasks.

There will also be a make-up assignment (based on last year's Assignment 1) for those who get catastrophically stuck, though this will usually be enough to get a passing grade only.

Baselines

Simple baselines have been provided for each of the tasks. These are included in 'baselines.py' (or ipynb) file among the files above. They are mostly intended to demonstrate how the data is processed and prepared for submission to gradescope.

- **Latent factor model:** For rating prediction tasks. Timestamps are ignored.
- **Bayesian Personalized Ranking:** For interaction prediction tasks. Timestamps / ordering is ignored.
- **Always predict median:** For time delta prediction

Running 'baselines.py' produces files containing predicted outputs (these outputs can be uploaded to gradescope). Your submission files should have the same format.